Question:
A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?
Answer Choices:
A. It activates TLR5 on the surface of macrophages.
B. It primarily induces the Th1-cell response.
C. Peptidoglycan is its major antigen that induces an immune response.
D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<tool_call> reasoning process here
The patient is a 34-year-old poultry worker with a history of sore throat, non-productive cough, fever, oropharyngeal erythema, and pulmonary findings consistent with community-acquired pneumonia (CAP). The chest X-ray shows patchy reticular opacities in the perihilar regions, which is typical of a bacterial infection, particularly one caused by a gram-negative bacterium commonly found in the environment, such as Haemophilus influenzae, Streptococcus pneumoniae, or even Legionella. However, the occupational history (poultry worker) raises suspicion for a pathogen commonly found in poultry environments, such as *Mycoplasma pneumoniae* or *Chlamydia pneumoniae*, or more specifically, *Klebsiella* or *Streptococcus* species.

However, the key clue is the presence of moist rales, perihilar opacities, and the fact that the patient is responding to cephalexin, a broad-spectrum antibiotic effective against Gram-positive bacteria and some Gram-negative bacteria. Cephalexin is particularly effective against *Streptococcus pneumoniae*, a common cause of community-acquired pneumonia.

Now, the question asks about the immune response elicited by the pathogen.

LetâĢĻs evaluate the options:

A. It activates TLR5 on the surface of macrophages.
TLR5 recognizes flagellin, a component of bacterial flagella. This is more relevant to flagellated bacteria such as *Salmonella*, *E. coli*, etc. Not a primary feature of common CAP pathogens like *S. pneumoniae* or *H. influenzae*, which lack flagella. So, not the best description.

B. It primarily induces the Th1-cell response.
Th1 responses are associated with intracellular pathogens like *Mycobacterium tuberculosis* or viruses. In bacterial pneumonia, especially from encapsulated bacteria like *S. pneumoniae*, the immune response involves both Th1 and Th17 responses, but the dominant innate response involves alveolar macrophages and neutrophils. However, *S. pneumoniae* is known to induce a strong Th17 response, which is more relevant to neutrophilic inflammation and lung tissue damage. Th1 is not the primary response. So, not the best choice.

C. Peptidoglycan is its major antigen that induces an immune response.
Peptidoglycan is a component of bacterial cell walls, especially in Gram-positive bacteria like *Streptococcus pneumoniae*. It is recognized by Toll-like receptors (e.g., TLR2 and TLR6) on macrophages and dendritic cells. This is a key component of the immune response to bacterial pneumonia. The pathogen causing this patientâĢĻs pneumonia is likely a Gram-positive bacterium (e.g., *S. pneumoniae*), and peptidoglycan is a major antigen that triggers the immune response. This is a correct and fundamental immunological fact.

D. This pathogen evades the immune response by encapsulation.
Encapsulation is a feature of *Streptococcus pneumoniae* and *Haemophilus influenzae*, which helps evade phagocytosis. However, the question asks about the immune response elicited, not evasion mechanisms. While encapsulation is true for some pathogens, it is not the best description of the immune response itself. The immune response is not defined by evasion but by antigen recognition